

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{50} + \frac{1}{50}$$

$$\frac{1}{f} = \frac{2}{50}$$

$$\frac{1}{f} = \frac{1}{25}$$

$$f = 25 \text{ cm}$$

$$\text{Power of lens} \Rightarrow P = \frac{1}{f(\text{in m})} = \frac{1}{25 \times 10^{-2}} = 4 \text{ m}^{-1}$$

$$10 = 1 \text{ m}^{-1}$$

$$4 \text{ m}^{-1} = 4 \text{ D}$$

∴ Object distance from lens is 50 cm and power of lens is 4 D.

Physics

① convex lens

② 10 cm

③ $u = 5\text{ cm}$ $f = ?$
 $v = 10\text{ cm}$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = -\frac{1}{10}$$

$$\frac{1}{f} = \frac{1}{10} - \frac{1}{5}$$

$$10 = -1f$$

$$-10 = f$$

$$\frac{1}{f} = \frac{1-2}{10}$$

focal length is -10 cm

Type of lens is concave.

④ $v = 50\text{ cm}$

height of image = h_i

height of object = h_o

} Given

magnification = -1

To find

$u = ?$

$P = ?$

$$m = \frac{v}{u}$$

$$-1 = \frac{50}{u}$$

$$u = -50\text{ cm}$$